

Lema de Riemann-Lebesgue en $\mathcal{L}^1(\mathbb{R})$

Lema 1 (de Riemann-Lebesgue). Sea $f \in \mathcal{L}^1(\mathbb{R})$, entonces $\lim_{|\xi| \rightarrow \infty} |f^\wedge(\xi)| = 0$.

Demostración: Aplicamos la identidad de Euler $e^{-i\pi} = -1$:

$$f^\wedge(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx = - \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} e^{-i\pi} dx = - \int_{\mathbb{R}} f(x) e^{-2\pi i (x + \frac{1}{2\xi}) \xi} dx.$$

Ahora hacemos el cambio de variable $y = x + \frac{1}{2\xi}$, con lo que

$$\begin{aligned} f^\wedge(\xi) &= - \int_{\mathbb{R}} f\left(y - \frac{1}{2\xi}\right) e^{-2\pi i y \xi} dy. \\ \implies 2f^\wedge(\xi) &= \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx - \int_{\mathbb{R}} f\left(y - \frac{1}{2\xi}\right) e^{-2\pi i y \xi} dy \\ \implies f^\wedge(\xi) &= \frac{1}{2} \left[\int_{\mathbb{R}} \left(f(x) - T_{\frac{1}{2\xi}} f(x)\right) e^{-2\pi i x \xi} dx \right] \end{aligned}$$

Por tanto, al calcular la norma, llegamos a que

$$|f^\wedge(\xi)| \leq \frac{1}{2} \int_{\mathbb{R}} \left| f(x) - T_{\frac{1}{2\xi}} f(x) \right| dx = \frac{1}{2} \left\| f - T_{\frac{1}{2\xi}} f \right\|_1.$$

Por el `lem-convergencia-lp-traslacion`/Lema 1, concluimos que $|f^\wedge(\xi)| \xrightarrow{|\xi| \rightarrow \infty} 0$. ■

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